

Strong normalization of the simply typed λ -calculus

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In this note, we review a proof of strong normalization of the simply typed λ -calculus, based on reducibility.

Definition 1 (Simply typed λ calculus). The set of types is given by:

$$A ::= \alpha \mid A \rightarrow A$$

Typing rules are given by:

$$\frac{}{\Gamma, x : A \vdash x^A : A} \text{ax} \quad \frac{\Gamma, x : A \vdash t : B}{\Gamma \vdash \lambda x^A. t : A \rightarrow B} \rightarrow \text{I} \quad \frac{\Gamma \vdash t : A \rightarrow B \quad \Gamma \vdash s : A}{\Gamma \vdash t s : B} \rightarrow \text{E}$$

Note that typable terms have unique type. Sometimes we do not write the type decoration on variables if it is clear from the context.

Note that the type of a term is unique (given that variables are decorated with their type), so sometimes we write $t : A$ to mean that t is a term of type A under some context. Reduction is given by the β -reduction rule:

$$(\lambda x. t) s \rightarrow_{\beta} t\{x := s\}$$

closed under arbitrary contexts. We write $\text{SN}(A)$ for strongly normalizing terms of type A . If $t \in \text{SN}(A)$, the length of reductions starting on t is bounded by König's lemma, as the λ -calculus is finitely branching. In that case, we write $|t|$ for the length of the longest reduction starting on t .

Definition 2 (Reducible terms). The set of reducible terms of type A is written $\llbracket A \rrbracket$ and defined as follows:

$$\llbracket A_1 \rightarrow \dots \rightarrow A_n \rightarrow \alpha \rrbracket = \{t : A_1 \rightarrow \dots \rightarrow A_n \rightarrow \alpha \mid \text{if } s_1 \in \llbracket A_1 \rrbracket, \dots, s_n \in \llbracket A_n \rrbracket \text{ then } t s_1 \dots s_n \in \text{SN}(\alpha) \}$$

Lemma 3 (Adequacy of variables). *The following hold:*

1. For all A and all x^A , we have $x^A \in \llbracket A \rrbracket$.
2. $\llbracket A \rrbracket \subseteq \text{SN}(A)$

Proof. We prove items 1. and 2. by simultaneous induction on A . Note that A can always be written as of the form $A = (A_1 \rightarrow \dots \rightarrow A_n \rightarrow \alpha)$. We prove each item:

1. To prove the first item, let $t_1 \in \llbracket A_1 \rrbracket, \dots, t_n \in \llbracket A_n \rrbracket$, and let us show that $x t_1 \dots t_n \in \text{SN}(\alpha)$. It suffices to show that $t_i \in \text{SN}(A_i)$. By the second item of the IH, it suffices to note that $t_i \in \llbracket A_i \rrbracket$.
2. To prove the second item, let $t \in \llbracket A_1 \rightarrow \dots \rightarrow A_n \rightarrow \alpha \rrbracket$. By IH note that $x_1^{A_1} \in \llbracket A_1 \rrbracket, \dots, x_n^{A_n} \in \llbracket A_n \rrbracket$. By definition, $t x_1^{A_1} \dots x_n^{A_n} \in \text{SN}(\alpha)$. In particular, $t \in \text{SN}(\alpha)$.

□

Lemma 4 (Adequacy of application). *If $t \in \llbracket A \rightarrow B \rrbracket$ and $s \in \llbracket A \rrbracket$ then $t s \in \llbracket B \rrbracket$.*

Proof. Let $t \in \llbracket A \rightarrow B \rrbracket$ and $s \in \llbracket A \rrbracket$. To show that $ts \in \llbracket B \rrbracket$, note first that B can always be written as of the form $B = B_1 \rightarrow \dots \rightarrow B_n \rightarrow \alpha$. Consider terms $u_1 \in \llbracket B_1 \rrbracket, \dots, u_n \in \llbracket B_n \rrbracket$, and let us show that $tsu_1 \dots u_n \in \text{SN}(\alpha)$. This is an immediate consequence of the fact that $t \in \llbracket A \rightarrow B_1 \rightarrow \dots \rightarrow B_n \rightarrow \alpha \rrbracket$. $\square$

Lemma 5 (Adequacy of abstraction). *If $t\{x := s\} \in \llbracket B \rrbracket$ holds for every $s \in \llbracket A \rrbracket$ then $\lambda x. t \in \llbracket A \rightarrow B \rrbracket$.*

Proof. First note that B can always be written as of the form $B = (B_1 \rightarrow \dots \rightarrow B_n \rightarrow \alpha)$. Suppose that $t\{x := s\} \in \llbracket B \rrbracket$ holds for every $s \in \llbracket A \rrbracket$. To show that $\lambda x. t \in \llbracket A \rightarrow B \rrbracket = \llbracket A \rightarrow B_1 \rightarrow \dots \rightarrow B_n \rightarrow \alpha \rrbracket$, let $s \in \llbracket A \rrbracket, u_1 \in \llbracket B_1 \rrbracket, \dots, u_n \in \llbracket B_n \rrbracket$, and let us show that $(\lambda x. t) s u_1 \dots u_n \in \text{SN}(\alpha)$. By hypothesis, we have that $t = t\{x := x\} \in \llbracket B \rrbracket$, hence by Lem. 3 $t \in \text{SN}(B)$. Moreover, $s, u_1, \dots, u_n \in \text{SN}$ by Lem. 3. Hence $|t|, |s|, |u_1|, \dots, |u_n|$ are defined. We proceed by induction on $|t| + |s| + |u_1| + \dots + |u_n|$. To show that $(\lambda x. t) s u_1 \dots u_n \in \text{SN}(\alpha)$, it suffices to show that whenever there is a β -step $(\lambda x. t) s u_1 \dots u_n \rightarrow_\beta t'$, the reduct is in turn strongly normalizing, i.e. $t' \in \text{SN}(\alpha)$. We consider four cases, depending on the position of the β -step, the interesting case is the last one:

1. Internal to t , that is, $(\lambda x. t) s u_1 \dots u_n \rightarrow_\beta (\lambda x. t') s u_1 \dots u_n$ where $t \rightarrow_\beta t'$. Then note that $|t| > |t'|$ so we may conclude by resorting to the IH.
2. Internal to s , that is, $(\lambda x. t) s u_1 \dots u_n \rightarrow_\beta (\lambda x. t) s' u_1 \dots u_n$ where $s \rightarrow_\beta s'$. Then note that $|s| > |s'|$ so we may conclude by resorting to the IH.
3. Internal to u_i for some $i \in 1..n$, that is,

$$(\lambda x. t) s u_1 \dots u_{i-1} u_i u_{i+1} \dots u_n \rightarrow_\beta (\lambda x. t) s u_1 \dots u_{i-1} u'_i u_{i+1} \dots u_n$$

where $u_i \rightarrow_\beta u'_i$. Then note that $|u_i| > |u'_i|$ so we may conclude by resorting to the IH.

4. Head reduction, that is, $(\lambda x. t) s u_1 \dots u_n \rightarrow_\beta t\{x := s\} u_1 \dots u_n$. Then, since $s \in \llbracket A \rrbracket$, by hypothesis, we have that $t\{x := s\} \in \llbracket B \rrbracket$, so by definition $t\{x := s\} u_1 \dots u_n \in \text{SN}(\alpha)$. $\square$

Definition 6 (Adequate substitution). A substitution σ is a function mapping variables to λ -terms. We write t^σ for the capture-avoiding substitution of all the free occurrences of each variable x in t by $\sigma(x)$. If $\Gamma = (x_1 : A_1, \dots, x_n : A_n)$, we say that a substitution is *adequate* for Γ , written $\sigma \models \Gamma$, if and only if $\sigma(x_i) \in \llbracket A_i \rrbracket$ for all $i \in 1..n$.

Theorem 7 (Adequacy). *Let $\Gamma \vdash t : A$ and let $\sigma \models \Gamma$. Then $t^\sigma \in \llbracket A \rrbracket$.*

Proof. By induction on the derivation of $\Gamma \vdash t : A$:

1. ax: Let $\Gamma, x : A \vdash x^A : A$. Then by adequacy of σ we have that $\sigma(x) \in \llbracket A \rrbracket$, as required.
2. $\rightarrow$ I: Let $\Gamma \vdash \lambda x. t : A \rightarrow B$ be derived from $\Gamma, x : A \vdash t : B$. Let $\sigma \models \Gamma$. To show that $(\lambda x. t)^\sigma \in \llbracket A \rightarrow B \rrbracket$, by Lem. 5, it suffices to show that for every $s \in \llbracket A \rrbracket$ we have that $t^\sigma \{x := s\} \in \llbracket B \rrbracket$. Define the substitution σ' as follows:

$$\sigma'(y) = \begin{cases} s & \text{if } x = y \\ \sigma(y) & \text{otherwise} \end{cases}$$

Note that $\sigma' \models \Gamma, x : A$. By IH, $t^{\sigma'} \in \llbracket B \rrbracket$. Finally, note that $t^\sigma \{x := s\} = t^{\sigma'}$, so we are done.

3. $\rightarrow$ E: Let $\Gamma \vdash ts : B$ be derived from $\Gamma \vdash t : A \rightarrow B$ and $\Gamma \vdash s : A$. By IH, we have that $t \in \llbracket A \rightarrow B \rrbracket$ and $s \in \llbracket A \rrbracket$. Hence by Lem. 4 we have that $ts \in \llbracket B \rrbracket$. $\square$

Corollary 8. *All typable terms are SN.*

Proof. Let $\Gamma \vdash t : B$. Let id denote the identity substitution. By Lem. 3 we have that $id \models \Gamma$, given that if $(x : A) \in \Gamma$ then $id(x) = x \in \llbracket A \rrbracket$. Hence, by Thm. 7, $t \in \llbracket B \rrbracket$. Moreover, by Lem. 3, $t \in \text{SN}(B)$. $\square$